

Section 2.4

46. $P(A|B) = \frac{P(A \cap B)}{P(B)}$. $P(B|A) = \frac{P(A \cap B)}{P(A)}$

Generally, the probability of an individual is over 6 ft in height is much larger than he/she is a professional basketball player. That is, $P(A) > P(B)$, $P(A|B) > P(B|A)$

Or, we know if an individual is a professional basketball player, he/she is much possible over 6 ft in height whereas the reverse is not so.

Thus $P(A|B) > P(B|A)$.

50. Let A_1, A_2, A_3 denote the event that the shirt sold is small, medium, large. Let B_1, B_2, B_3 denote plaid, print and stripe three patterns.

And C_1, C_2 denote long and short two sleeve lengths.

a. Based on the tables, we can know

$$P(A_2 \cap B_2 \cap C_1) = 0.05$$

$$\begin{aligned} \text{b. } P(A_2 \cap B_2) &= P(A_2 \cap B_2 \cap C_1) + P(A_2 \cap B_2 \cap C_2) \\ &= 0.07 + 0.05 \\ &= 0.12 \end{aligned}$$

$$c. P(C_2) = P(A_1 \cap C_2) + P(A_2 \cap C_2) + P(A_3 \cap C_2)$$

$$= P(A_1 \cap C_2 \cap B_1) + P(A_1 \cap C_2 \cap B_2) + P(A_1 \cap C_2 \cap B_3) \\ + P(A_2 \cap C_2 \cap B_1) + P(A_2 \cap C_2 \cap B_2) + P(A_2 \cap C_2 \cap B_3) \\ + P(A_3 \cap C_2 \cap B_1) + P(A_3 \cap C_2 \cap B_2) + P(A_3 \cap C_2 \cap B_3)$$

$$= 0.04 + 0.08 + 0.03 + 0.02 + 0.07 + 0.07 + 0.05 + 0.12 \\ + 0.08$$

$$= 0.56$$

$$P(C_1) = 1 - P(C_2) = 0.44$$

$$d. P(A_2) = P(A_2 \cap C_1) + P(A_2 \cap C_2)$$

$$= P(A_2 \cap C_1 \cap B_1) + P(A_2 \cap C_1 \cap B_2) + P(A_2 \cap C_1 \cap B_3) \\ + P(A_2 \cap C_2 \cap B_1) + P(A_2 \cap C_2 \cap B_2) + P(A_2 \cap C_2 \cap B_3)$$

$$= 0.10 + 0.05 + 0.07 + 0.08 + 0.07 + 0.12 = 0.49$$

$$P(B_2) = P(B_2 \cap C_1) + P(B_2 \cap C_2)$$

$$= P(B_2 \cap C_1 \cap A_1) + P(B_2 \cap C_1 \cap A_2) + P(B_2 \cap C_2 \cap A_3) \\ + P(B_2 \cap C_2 \cap A_1) + P(B_2 \cap C_2 \cap A_2) + P(B_2 \cap C_2 \cap A_3)$$

$$= 0.02 + 0.05 + 0.02 + 0.02 + 0.07 + 0.07$$

$$= 0.25$$

$$e. P(A_2 | B_1 \cap C_2) = \frac{P(A_2 \cap B_1 \cap C_2)}{P(B_1 \cap C_2)}$$

$$P(B_1 \cap C_2) = P(A_1 \cap B_1 \cap C_2) + P(A_2 \cap B_1 \cap C_2) + P(A_3 \cap B_1 \cap C_2)$$

$$= 0.04 + 0.08 + 0.03$$

$$= 0.15$$

$$P(A_2 \cap B_1 \cap C_2) = 0.08$$

$$\text{Thus } P(A_2 | B_1 \cap C_2) = \frac{0.08}{0.15} = 0.533$$

$$f. P(C_2 | A_2 \cap B_1) = \frac{P(A_2 \cap B_1 \cap C_2)}{P(A_2 \cap B_1)}$$

$$P(A_2 \cap B_1) = P(A_2 \cap B_1 \cap C_1) + P(A_2 \cap B_1 \cap C_2)$$

$$= 0.08 + 0.10$$

$$= 0.18$$

$$P(A_2 \cap B_1 \cap C_2) = 0.10$$

$$\text{Thus } P(C_2 | A_2 \cap B_1) = \frac{0.10}{0.18} = 0.556$$

$$P(C_1 | A_2 \cap B_1) = \frac{P(A_2 \cap B_1 \cap C_1)}{P(A_2 \cap B_1)}$$

$$P(A_2 \cap B_1 \cap C_1) = 0.08$$

$$\text{Thus } P(C_1 | A_2 \cap B_1) = \frac{0.08}{0.18} = 0.444$$

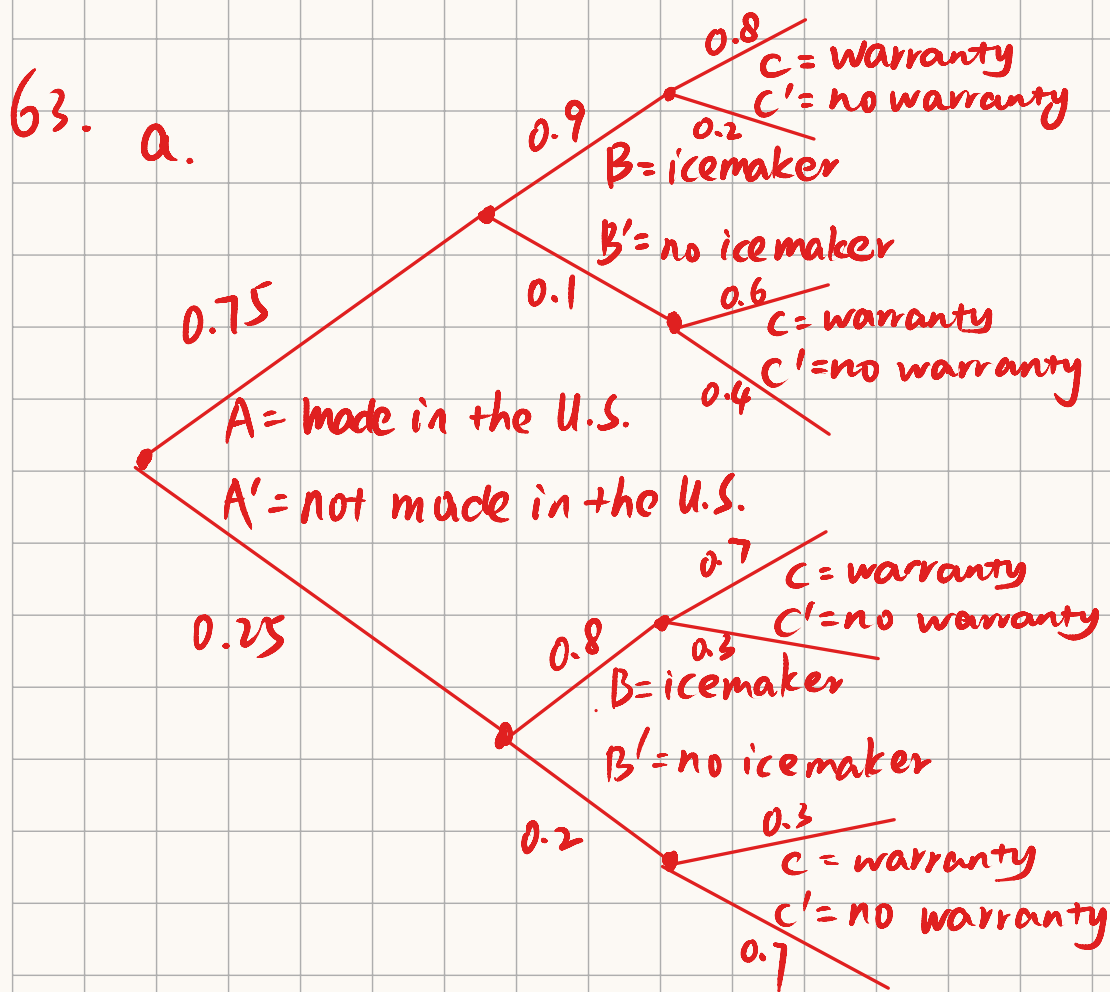
58.
$$P(A \cup B | C) = \frac{P((A \cup B) \cap C)}{P(C)}$$

$$= \frac{P((A \cap C) \cup (B \cap C))}{P(C)} \quad (\text{De Morgan's Law})$$

$$= \frac{P(A \cap C) + P(B \cap C) - P(A \cap B \cap C)}{P(C)} \quad (\text{Inclusive-Exclusive principle})$$

$$= \frac{P(A \cap C)}{P(C)} + \frac{P(B \cap C)}{P(C)} - \frac{P(A \cap B \cap C)}{P(C)}$$

$$= P(A|C) + P(B|C) - P(A \cap B|C)$$



$$\begin{aligned}
 \text{b. } P(A \cap B \cap C) &= P(C | A \cap B) \cdot P(B | A) \cdot P(A) \\
 &= 0.75 \times 0.9 \times 0.8 \\
 &= 0.54
 \end{aligned}$$

$$\begin{aligned}
 \text{c. } P(B \cap C) &= P(A \cap B \cap C) + P(A' \cap B \cap C) \\
 P(A' \cap B \cap C) &= P(C | A' \cap B) \cdot P(B | A') \cdot P(A') \\
 &= 0.25 \times 0.8 \times 0.7 \\
 &= 0.14
 \end{aligned}$$

$$\text{Thus } P(B \cap C) = 0.54 + 0.14 = 0.68$$

$$\begin{aligned}
 \text{d. } P(C) &= P(A \cap C) + P(A' \cap C) \\
 &= P(A \cap B \cap C) + P(A \cap B' \cap C) \\
 &\quad + P(A' \cap B \cap C) + P(A' \cap B' \cap C)
 \end{aligned}$$

$$\begin{aligned}
 P(A \cap B' \cap C) &= P(C | A \cap B') \cdot P(B' | A) \cdot P(A) \\
 &= 0.75 \times 0.1 \times 0.6 \\
 &= 0.045
 \end{aligned}$$

$$\begin{aligned}
 P(A' \cap B' \cap C) &= P(C | A' \cap B') \cdot P(B' | A') \cdot P(A') \\
 &= 0.25 \times 0.2 \times 0.3 \\
 &= 0.015
 \end{aligned}$$

$$\text{Thus } P(C) = 0.54 + 0.045 + 0.14 + 0.015 = 0.74$$

$$e. P(A|B \cap C) = \frac{P(A \cap B \cap C)}{P(B \cap C)} = \frac{0.54}{0.68} = 0.794$$

